

Al buio non si trova

Biostatistics in the 21st century

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Available from: <https://github.com/maxbiostat/presentations/>



Plan for today

Searching in the dark

A little metaphor to get us in the mood

Problem I: using historical data to the fullest

Optimal Bayesian dynamic borrowing of information

Problem II: dealing with huge complex data

MCMC in tree space: a journey through a strange land

Where the magic happens

Academic life at FGV EMAP

A (useful?) methaphor

Che gelida manina (La Boheme, Puccini, 1896)

Che gelida manina, se la lasci riscaldar.

What a frozen little hand, let me warm it for you.

Cercar che giova? Al buio non si trova.

What's the use of looking? We won't find it in the dark.

Ma per fortuna, è una notte di luna,

But luckily, it's a moonlit night,

E qui la luna... l'abbiamo vicina.

and the moon is near us here.

Roberto Alagna & Leontina Vaduva, Paris, 1995.

Le dirò con due parole, chi son

Personal

- ⊙ Born and raised in Petrópolis-RJ;
- ⊙ Eldest of three kids;
- ⊙ Married and father of two sassy girls;
- ⊙ Mais Querido supporter.

Academic

- ⊙ BSc in Microbiology & Immunology (UFRJ, 2012);
- ⊙ PhD Evolutionary Biology (Edinburgh, 2018);
- ⊙ Post doctoral researcher at ENSP/Fiocruz (2019);
- ⊙ Lecturer (Assistant Professor) at EMAp since Jan/2020.

Applications of Statistics/Mathematics

Applications in Epidemiology, (Molecular) Biology, Ecology, Psychology, Linguistics, etc.

Applied Statistics

Markov Chain Monte Carlo, Model combination and selection, Statistical Phylogenetics.

Problem I: efficiently utilising available information

Loads of historical data: how to build informative priors?

Let $\mathbf{y}_0 = (y_{01}, \dots, y_{0n_0})$ and $\mathbf{y} = (y_1, \dots, y_n)$ be **historical** and **current** data, respectively.

Question: how do I build a prior that

- ⊙ Uses information in $\mathbf{y}_0$ efficiently but also
- ⊙ Does not lead to borrowing too much information when the data sets are not compatible?

Applications: clinical trials, quality control, policy-making.

Normalised power prior (NPP)¹

$$\tilde{\pi}(\theta, a_0 \mid \mathbf{y}_0) \propto \frac{L_0(\mathbf{y}_0 \mid \theta)^{a_0} \pi_0(\theta \mid \phi) \pi_A(a_0 \mid \xi)}{c(a_0; \phi)}$$

- ⊙ How pick π_A such that prediction error (say) is minimised?²
- ⊙ How to **efficiently** compute

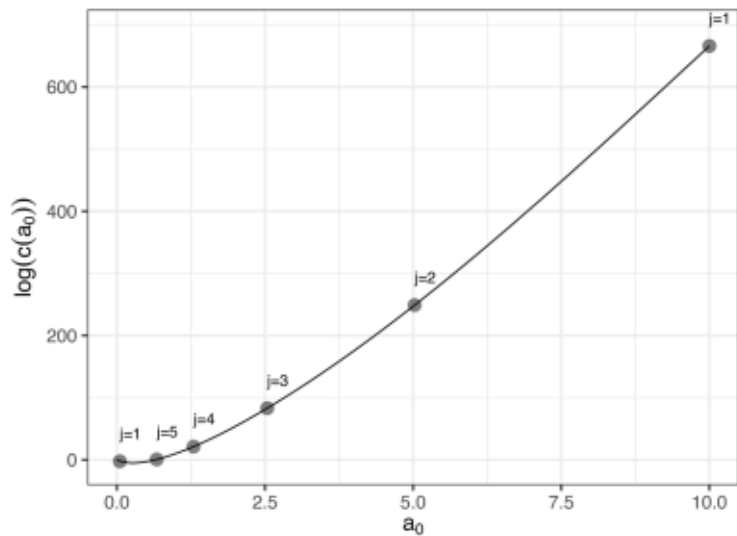
$$c(a_0; \phi) = \int_{\Theta} L_0(\mathbf{y}_0 \mid t)^{a_0} \pi_0(t \mid \phi) d\mu(t)$$

by leveraging its special properties as function of a_0 ?

¹Carvalho & Ibrahim (2021) *Statistics in Medicine*.

²Shen et al. (2024) *Statistica Sinica*.

Approximating the normalising constant



What we are doing right now

Principled Bayesian analysis under the NPP³

Calibrate the initial prior π_0 and pick the prior on the discounting parameter a_0 *via* prior predictive checking.

Exact MCMC for the NPP⁴

The approximate version comes with no guarantees

Variational approximations for large data/models⁵

Sometimes approximate is all you have, but need provable performance

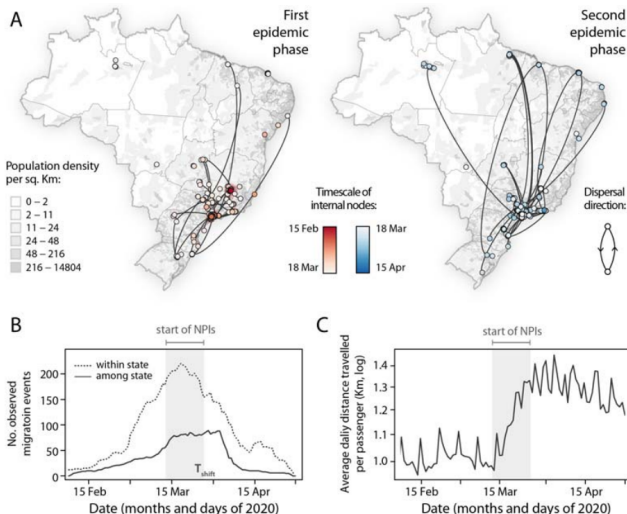
³Ezequiel Braga (MSc)

⁴Eduardo Adame (MSc)

⁵Yure Oliveira (IC)

Problem II: dealing with huge complex data

Where did this virus come from?⁶



⁶Cândido et al. (2020) *Science*.

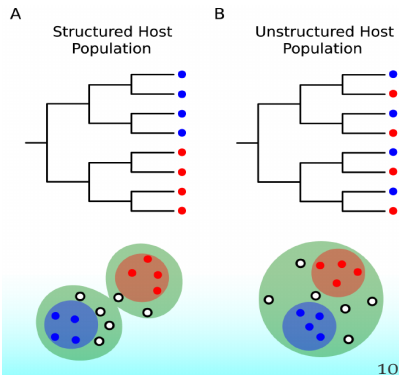
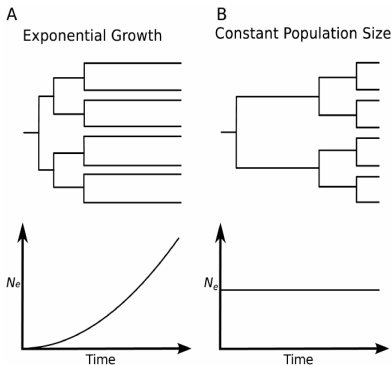
Motivation

Phylogenetics of fast-evolving viruses

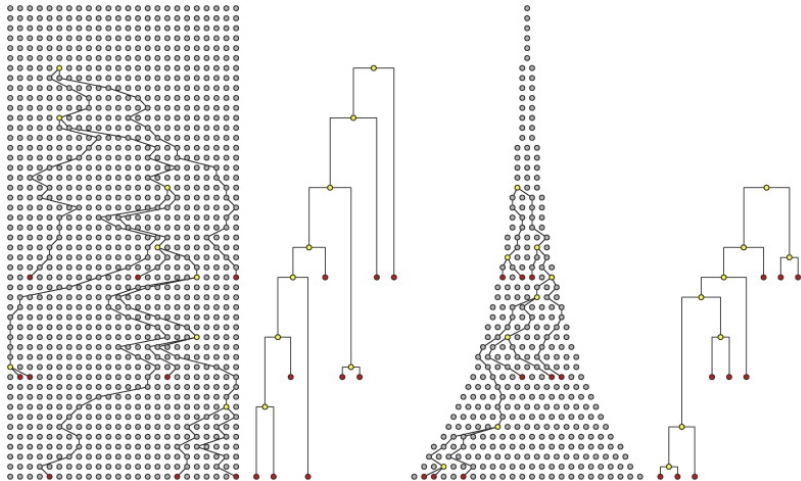
Inferring spatial and temporal dynamics from genomic data:

Phylogenies*!

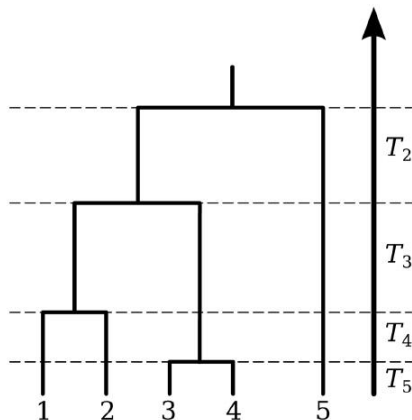
* plus complicated models



Trees and the coalescent



Central object: time-calibrated trees



Let T_n denote the time for n lineages to *coalesce*, i.e., merge into one ancestral lineage, in a population of size N_e . Then:

$$\Pr(T_n = t) = \lambda_n e^{-\lambda_n t}$$

$$\lambda_n = \binom{n}{2} \frac{1}{N_e} = \binom{n}{2} \frac{1}{N_e \tau}$$

where N_e is the effective population size and τ is the generation time. Let T_{mrca} denote the age of the most recent common ancestor:

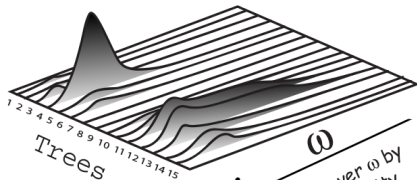
$$\begin{aligned} \mathbb{E}[T_{\text{mrca}}] &= \mathbb{E}[T_n] + \mathbb{E}[T_{n-1}] + \dots + \mathbb{E}[T_2] \\ &= 1/\lambda_n + 1/\lambda_{n-1} + \dots + 1/\lambda_2 \\ &= 2N_e \left(1 - \frac{1}{n}\right) \end{aligned}$$

Figure: Figure 4 from [Volz et al. \(2013\)](#).

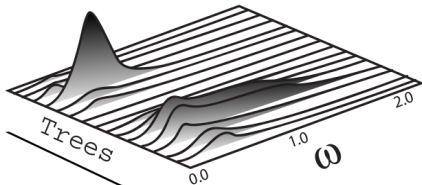
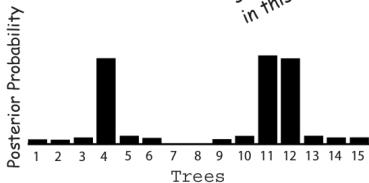
$$p(t, \mathbf{b}, \boldsymbol{\omega} | D) = \frac{f(D | t, \mathbf{b}, \boldsymbol{\omega}) \pi(t, \mathbf{b}, \boldsymbol{\omega})}{\sum_{t_i \in T_n} \int_B \int_{\Omega} f(D | t_i, \mathbf{b}_i, \boldsymbol{\omega}) \pi(t_i, \mathbf{b}_i, \boldsymbol{\omega}) d\boldsymbol{\omega} d\mathbf{b}_i} \quad (1)$$

- ⊙ D : observed sequence (DNA) data;
- ⊙ T_n : set of all binary ranked trees;
- ⊙ $\mathbf{b}_k$: set of branch lengths of $t_k \in T_n$ ($\mathbb{R}_+^{2n-2}$, kind of) ;
- ⊙ $\boldsymbol{\omega}$: set of parameters of interest such as substitution model parameters, migration rates, heritability coefficients, etc.

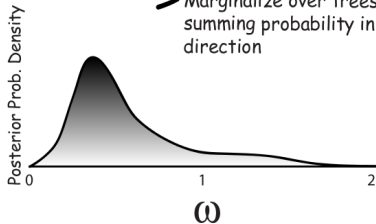
The end product



← Marginalize over ω by summing probability in this direction

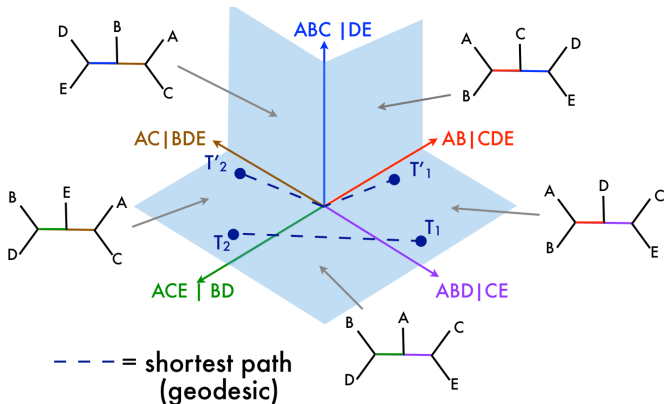


→ Marginalize over trees by summing probability in this direction



This place is weird..

Traversing cubic complexes efficiently⁷



Applications: Molecular Epidemiology, Evolutionary Biology.

⁷<https://youtu.be/h9bWRQ6aeKA>

(Adaptive) Metropolis-Hastings for trees

General MH setup.

Let $\tau = (t, \mathbf{b})$ denote a tree with topology t and branch lengths $\mathbf{b}$. For two trees τ and τ' , denote the transition kernel by $q_\gamma(\tau|\tau') := \Pr(\tau' \rightarrow \tau|\gamma)$.

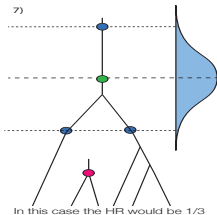
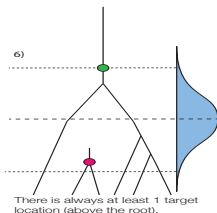
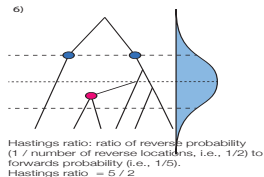
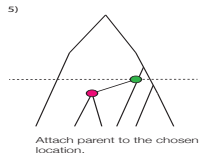
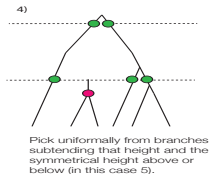
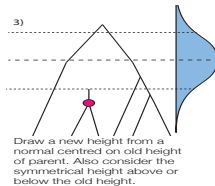
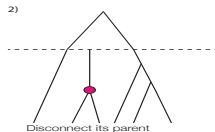
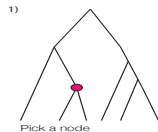
Accepting with probability

$$A_\gamma(\tau|\tau') = \min \left(1, \frac{p(\tau', \boldsymbol{\omega}|D)q_\gamma(\tau|\tau')}{p(\tau, \boldsymbol{\omega}|D)q_\gamma(\tau'|\tau)} \right)$$

leads to the desired target.

Note: Here $\gamma > 0$ is a so-called tuning parameter.

STL – illustration



Carvalho (2019), Chapter 2.

Lemma

Assume strictly positive branch lengths. Then SubTreeLeap induces an irreducible Markov chain on T_n .

Sketch: Starting at $x \in T_n$, notice there exists $\delta_y^\star > 0$ such that $P(x \rightarrow y \mid \delta_y^\star) > 0$ for any tree $y \in T_n$ in the SPR neighbourhood of x .

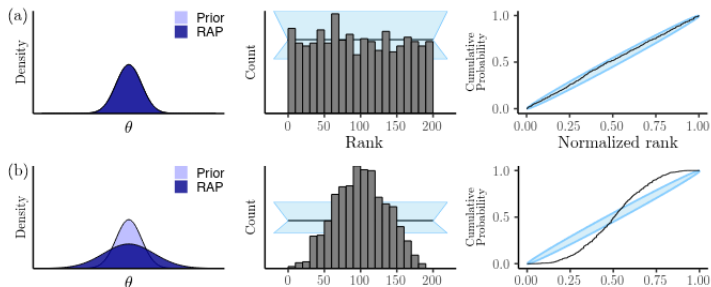
Theorem

Assume the target satisfies $p(A) > 0$ for all $A \in \Psi$. Then, SubTreeLeap induces an ergodic Markov chain on Ψ .

Sketch: Employ the lemma to get to the case where $d_{\text{SPR}}(x, y) = 0$ and then establish Harris recurrence.

How do you tell if it worked, though?

- ⊙ **Super important** to check consider actual tree metrics when checking convergence (convergence on ω is not informative)⁸.
- ⊙ **Validating** the methodology using rigorous modern techniques such as Simulation-based Calibration (SBC)⁹



⁸Brusselmans et al (2024) *Virus Evolution*

⁹Mendes et al. (2025) *Systematic Biology*

Open problems in MCMC for phylogenies

Open problems:

- Theoretical aspects of phylogenetic MCMC¹⁰;
- How can we construct more efficient proposals? How to exploit structure? **Geometry!**
- How to quantify exploration of the target? (Custom) **Tools!**
- Optimal scaling: what's the **optimal** acceptance probability?
- New, statistically-motivated metrics and measures of centrality (e.g. HIPSTR¹¹.)

¹⁰See [Semper & Palacios \(2022\)](#) and [Alves, Saporito & Carvalho \(2022\)](#) for recent attempts.

¹¹[Baele et al \(2025\) *Bioinformatics*](#).

Onde a mágica acontece – FGV EMaP



A ESCOLA

Excelência na aplicação da matemática e da computação

Com método de ensino participativo e professores altamente qualificados, a FGV EMaP prepara profissionais para desenvolverem matemática e modelos computacionais de forma inovadora, adaptados aos desafios da era da informação.

Aqui, os alunos saem com uma base sólida para atuar em setores estratégicos de organizações públicas e privadas, qualificando-os, também, para a realização de pesquisas acadêmicas e projetos de consultoria.

Conheça a escola >



- Mestrado e Doutorado em **Matemática Aplicada e Ciência de Dados**;
- Edital aberto até 30/9;
- Seleção: análise de currículo, e se aprovado, entrevista;
- **Bolsa de mestrado: R\$ 3.000,00;**
- **Bolsa de doutorado: R\$ 4.000,00;**
- Possibilidade de doutorado direto;
- Bolsas de projetos e monitoria.

Mestrado

- 7 disciplinas: 4 centrais e 3 eletivas
- 2 disciplinas de seminários

Doutorado

- 7 disciplinas: 3 centrais e 4 eletivas
- 4 disciplinas de seminários
- Exame de Qualificação I, II, III e Tese

■ Centrais do Doutorado (e Mestrado)

Medida, Integração e Probabilidade

Otimização

Teoria Qualitativa de EDO

Controle Ótimo

Análise Numérica e Simulação

Processos Estocásticos

Análise Funcional: Fundamentos

Equações Diferenciais Estocásticas: Teoria e Simulação

Análise Funcional: Operadores Lineares

Finanças Quantitativas

Ciência de Redes

Métodos Quantitativos em Risco

Equações Diferenciais Parciais e Aplicações

Biologia Matemática

Visualização para Aprendizado de Máquinas

Causalidade

Aprendizado de Máquinas

Epidemiologia Matemática

Modelos Generativos

Inferência Estatística

Redes Neurais e Aprendizado Profundo

Estatística Computacional

Aprendizado por Reforço

Estatística Bayesiana

Nossos egressos



Lucas Moschen
Imperial College
London (antes,
Sorbonne)



Lucas Resck
Cambridge
University



Vitória Guardieiro
University of
Pennsylvania

Take home

A light in the dark

Scalable methods with provable guarantees

Computational methods are key

Learn to program and learn Computational Statistics¹²

We've got loads to do!

Biomedical statistics is where most of the cool data and problems **with actual impact** are.

Come work with us!

Come to the School of Applied Mathematics (FGV EMaP)

¹²Here's a place to start:

https://github.com/maxbiostat/Computational_Statistics

THE
END